Al Imam Mohammad Ibn Saud Islamic University
College of Computer and Information Sciences
Computer Science Department

Course Title:	Design and Analysis of Algorithms
Course Code:	CS216
Course Instructors:	Dr. Dr.Yassine Daadaa and Dr. Asma Al-Saleh
Exam:	Midterm 2 (MODEL ANSWER)
Semester:	Second Semester
Date:	4/4/2021 (4-5 pm)
Duration:	60 minutes
Marks:	20 Marks
Privileges:	☐ ~~Open Book~~ ☐ ~~Open Notes~~ ☐ Calculator Permitted ☐ ~~Laptop Permitted~~

Student Name (in English):	
Student ID:	
Section No.:	

Instructions:

1. Answer all questions. There are 4 questions in 7 pages.
2. Write your name on each page of the exam paper.
3. Write your answers directly on the question sheets. Use the ends of the question pages for rough work or if you need extra space for your answer.
4. If information appears to be missing from a question, make a reasonable assumption, state your assumption, and proceed.

Official Use Only		
Question	**Student Marks**	**Question Marks**
1		6
2		7
3		5
4		2
Total		**20**

Student Name (in English): _____ **Student ID:** _____

Question 1: **Multiple choices** To be answered in **(15) Minutes** [ ] / 6 **Marks**

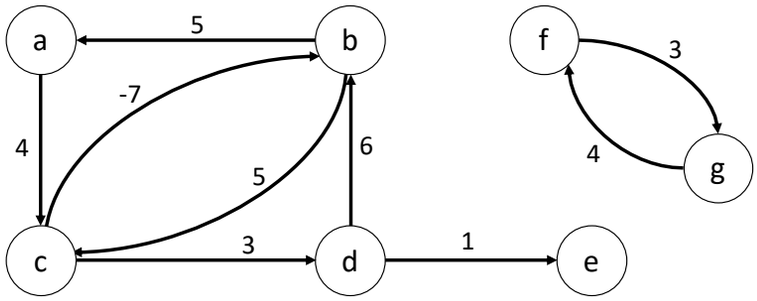
Question	1	2	3	4	5	6
Answer	c	b	c	b	d	b

1. **Consider the list of integers: 7, 107, 408, 210, 15, 329, 141, 53, 663, the output of the list after the *first pass of* the *radix* sort algorithm?**
   - **a)** 7, 15, 53, 107, 141, 210, 329, 408, 663
   - **b)** 7, 107, 408, 210, 15, 329, 141, 53, 663
   - **c)** 210, 141, 53, 663, 15, 7, 107, 408, 329
   - **d)** All the above the answers are incorrect

2. **In *radix* sort, to sort binary numbers, ____ buckets are needed.**
   - **a)** 1
   - **b)** 2
   - **c)** 3
   - **d)** 10

3. **We can apply _____ sort on any list of items.**
   - **a)** Counting
   - **b)** Bucket
   - **c)** Merge
   - **d)** linear

Consider this the following graph and answer the remaining questions (4, 5, and 6).



4. **The order in which vertices are *discovered and fully explored* in the Depth First Search starting from vertex a is: (Note: use alphabetical order to break ties)**
   - **a)** b→e→d→c→a
   - **b)** b→e→d→c→a→g→f
   - **c)** a→c→b→d→e→f→g
   - **d)** All the above answers are incorrect

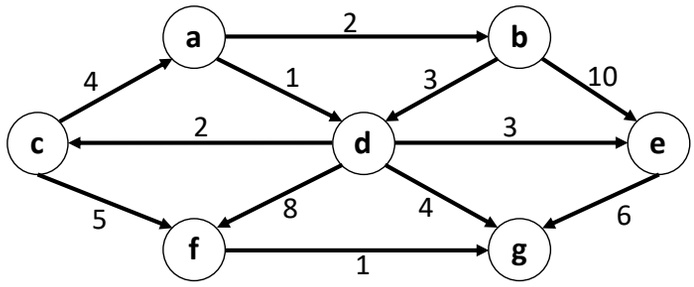
5. **To find the shortest path from vertex a to each other vertex in the above graph, we can used:**
   - **a)** Dijkstra's algorithm
   - **b)** Bellman-ford Algorithm
   - **c)** Both answers (a) and (b) are correct
   - **d)** All the above answers are incorrect

6. **To pass through *each* vertex in the above graph, we can use:**
   - **a)** Breadth first search algorithm
   - **b)** Depth first search algorithm
   - **c)** Both answers (a) and (b) are correct
   - **d)** Both answers (a) and (b) are incorrect

Student Name (in English): _____ **Student ID:** _____

Question 2 (A): **Graph** To be answered in **(20) Minutes** [ ] / 7 **Marks**

A. Consider the following graph: **(4 marks)**



i. Run Dijkstra's algorithm on the following directed graph, starting at the source vertex ***a*** and fill the table below (Note: **d** is the *total distance* from the source vertex to each vertex in the graph and π is the *predecessor* vertex of each vertex). (3.5 points)

Vertex	a	b	c	d	e	f	g
d	0	2	3	1	4	8	5
π	nil	a	d	a	d	c	d

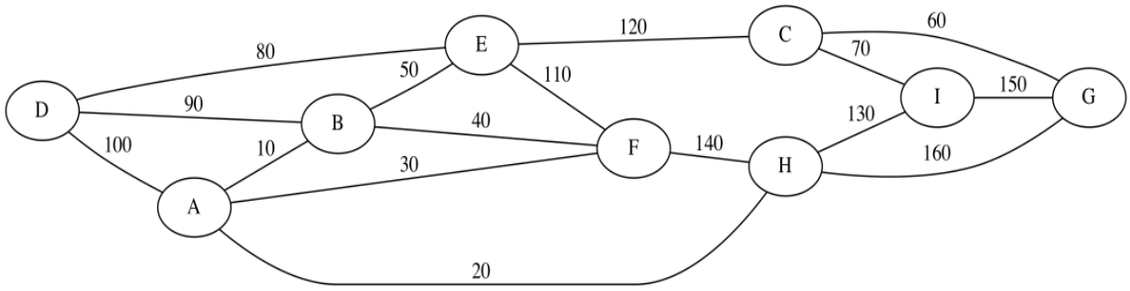
ii. What is the order in which vertices get removed from the priority queue? (0.5 points)

a→d→b→c→e→g→f

Student Name (in English): _____ **Student ID:** _____

Question 2 (B) Graph To be answered in **(20) Minutes** [ ] / 7 **Marks**

B. Consider the following weighted undirected graph: (3 marks)



i. Give the order in which the edges are added to the Minimum Spanning Tree using *Prim's algorithm*. Start at vertex A.

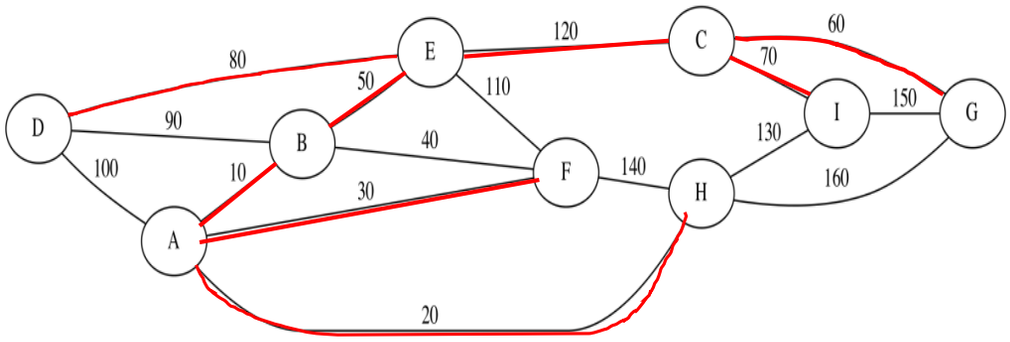
A-B→ A-H → A-F → B-E→ D-E → E-C →C-G → C-I

0.25 → each edge.

0.5 for total cost.

Order→ 0.5

ii. Draw obtained Minimum Spanning Tree.



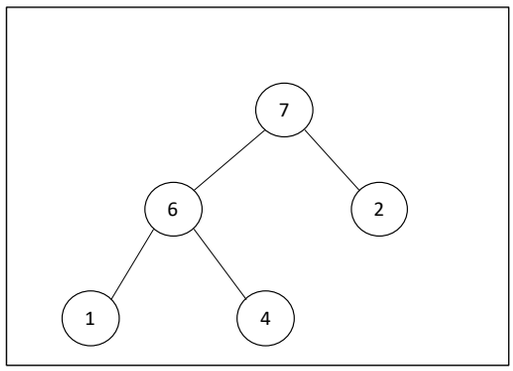
iii. What is the total cost of the tree? ___**440**___

Student Name (in English): _____ **Student ID:** _____

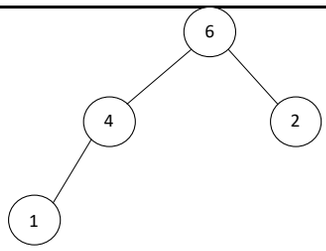
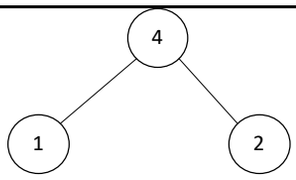
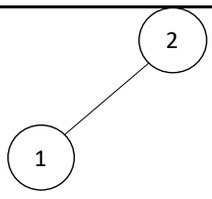
Question 3: **Sorting** To be answered in **(10) Minutes [ ] / 5 Marks**

A. Illustrate sorting the array A= {4, 6, 2, 1, 7} using *heap sort* assuming the max heap has been already built. First, draw the heap and then draw each step by showing the tree and the corresponding array. (5 points)

The heap: (1 point)



The heapsort step: (1 point for each step)

Step #	Tree	Array
Step 1		6 | 4 | 2 | 1 | 7
Step 2		4 | 1 | 2 | 6 | 7
Step 3		2 | 1 | 4 | 6 | 7
Step 4		1 | 2 | 4 | 6 | 7

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Question 4: To be answered in **(10) Minutes [ ] / 2 Marks**

Consider the Kruskal's algorithm below. (2 points)

```
MST-KRUSKAL(G, w)
1  A = ∅
2  for each vertex v ∈ G.V
3       MAKE-SET(v)
4  sort the edges of G.E into nondecreasing order by weight w
5  for each edge (u, v) ∈ G.E, taken in nondecreasing order by weight
6       if FIND-SET(u) ≠ FIND-SET(v)
7            A = A ∪ {(u, v)}
8            UNION(u, v)
9  return A
```

a) Modify the Kruskal's algorithm in more efficient way if we assume that the edge weights in the input graphs are uniformly distributed over [0, 1). What is the modification you will apply?

Use bucket sort to sort the edges (line 4).

b) What is the overall time complexity of the modified version of the algorithm. Show the details.

O(V+E+Elog v) or O(Elog v)